Semester 7													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24INT451T	Summer Internship	0	0	0	0	2	--	100	--	--	--	100
OE-4	Open Elective 4 (Inter-department, FoET)	3	0	0	3	3	25	25	50	--	--	100
24CH401T	Process Modelling and Optimization	3	0	0	3	3	25	25	50	--	--	100
24CH401P	Process Modelling and Optimization Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH402T	Transport Phenomena	3	1	0	4	4	25	25	50	--	--	100
24CH403T	Computer Aided Process Design	3	0	0	3	3	25	25	50	--	--	100
24CH403P	Computer Aided Process Design Laboratory	0	0	2	2	1	--	--	--	50	50	100
PE-4	Programme Elective 4	Students must choose one course from the given basket of elective courses for study										
24CH431T	Polymer Science & Technology	3	0	0	3	3	25	25	50	--	--	100
24CH432T	Energy Conversion Device Engineering											
24CH433T	Pharmaceuticals Technology											
PE-5	Programme Elective 5	Students must choose one course from the given basket of elective courses for study										
24CH434T	Process Plant Safety, Health and Hygiene	3	0	0	3	3	25	25	50	--	--	100
24CH435T	Project Management											
24CH436T	Plant Design & Process Economics											
24PRCH451T	Seminar	0	0	0	0	0	--	--	--	--	--	Audit
	Total	**18**	**1**	**6**	**23**	**23**						

Semester 8													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24PRCH452T/ 24PRCH453T	Major Project / Comprehensive Project	0	0	26	26	13	--	100	--	--	--	100
	Total	**0**	**0**	**26**	**26**	**13**						

Note: Students shall opt for any one of 24YOG101 / 24NSS101 / 24NCC101.
Note: 24INT251T Industrial Orientation is a mandatory audit (PP/NP) component carrying zero credits.
Note: Students shall register for any one of 24PRCH452T Major Project or 24PRCH453T Comprehensive Project.
Total credits for the award of the degree: 160

24PRCH452T					Major Project					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	26	13	26	--	--	100 (CE)	--	--	100

COURSE OBJECTIVES

- To apply the knowledge of chemical engineering to a substantial open-ended problem of industrial or research relevance.
- To develop the ability to plan, execute and manage an engineering project within realistic constraints.
- To develop competence in design, experimentation, modelling, simulation and analysis as applicable to the problem.
- To develop professional documentation, presentation and teamwork skills.

COURSE CONTENT / ACTIVITIES

1. Problem identification, literature survey and formulation of objectives and methodology.
2. Preparation and presentation of the project synopsis before the departmental committee.
3. Execution of the work: design, experimentation, modelling, simulation, fabrication or analysis as applicable.
4. Periodic progress reviews with the faculty guide and the departmental evaluation committee.
5. Analysis and interpretation of results, including consideration of safety, environment, ethics, economics and sustainability.
6. Submission of the project report in the prescribed format with a plagiarism check, followed by final presentation and viva-voce.

Max 338 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall and consolidate the core knowledge of chemical engineering relevant to the problem undertaken.
CO2 - Understand the state of the art through a systematic literature survey and define the problem precisely.
CO3 - Apply appropriate engineering, computational and experimental tools to address the problem.
CO4 - Analyse and interpret the results obtained, with due regard to safety, environment, ethics and sustainability.
CO5 - Evaluate the outcomes against the stated objectives and appraise the economic and societal implications.
CO6 - Design and communicate a complete engineering solution through a professional report and presentation while functioning effectively in a team.

TEXT / REFERENCE BOOKS

1. Pandit Deendayal Energy University. (2024). *Project guidelines and evaluation rubrics*. PDEU.
2. Day, R. A., & Gastel, B. (2016). *How to write and publish a scientific paper* (8th ed.). Cambridge University Press.

EVALUATION

Evaluation is by Continuous Evaluation (CE) for 100 marks, comprising the synopsis, periodic progress reviews, project report, final presentation and viva-voce assessed by the departmental committee and the external examiner.

24PRCH453T					Comprehensive Project					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	26	13	26	--	--	100 (CE)	--	--	100

COURSE OBJECTIVES

- To integrate and apply the entire body of chemical engineering knowledge to a comprehensive plant or process design problem.
- To develop the ability to carry out material and energy balances, equipment design and costing for a complete process.
- To develop competence in process simulation, safety analysis and economic evaluation.
- To develop professional documentation, presentation and teamwork skills.

COURSE CONTENT / ACTIVITIES

1. Selection of the process and comprehensive literature and technology survey, including market and capacity assessment.
2. Preparation of the process flow diagram, material and energy balances for the selected capacity.
3. Process design and detailed design of major equipment, with process simulation where applicable.
4. Preparation of the piping and instrumentation diagram, plant layout and utility requirements.
5. Hazard identification, safety and environmental appraisal of the proposed plant.
6. Cost estimation, profitability and sensitivity analysis; submission of the report with a plagiarism check, followed by final presentation and viva-voce.

Max 338 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall and consolidate the core knowledge of chemical engineering required for a complete process design.
CO2 - Understand the selected process technology through a comprehensive technology and literature survey.
CO3 - Apply material and energy balances, thermodynamics, transport and reaction engineering principles to the selected process.
CO4 - Analyse and design the major equipment of the plant along with the associated utilities and instrumentation.
CO5 - Evaluate the plant with respect to safety, environment, sustainability and economics.
CO6 - Design and communicate a complete plant design through a professional report and presentation while functioning effectively in a team.

TEXT / REFERENCE BOOKS

1. Pandit Deendayal Energy University. (2024). *Project guidelines and evaluation rubrics*. PDEU.

EVALUATION

Evaluation is by Continuous Evaluation (CE) for 100 marks, comprising the synopsis, periodic progress reviews, project report, final presentation and viva-voce assessed by the departmental committee and the external examiner.